

WEBFOSS-AT

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Work Package 1: Open Source Tools for URL Finding and Information Extraction, Statistical Scraping Interest Group (SSIG)

Deliverable 1.4: Report on SSIG3

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Content

1. Introduction
2. Participants
3. Agenda
4. Meeting summary
 - 4.1 Start, welcome, SSIG in brief, who is here
 - 4.2 Statistical scraping Use-cases in Austria: WEB-FOSS-AT / R-package taRantula
 - 4.3 Statistical Scraping in Netherlands: WEB-FOSS-NL
 - 4.4 Integrating Domain Data into the Danish Statistical Business Register
 - 4.5 An Atlas of Economic Activities in the UK
 - 4.6 Job title cleaning pipeline
 - 4.7 Recent Achievements in Project Implementation at Statistics Poland
 - 4.8 WebCAP: Web Data Collection Via Automated Program Synthesis
 - 4.9 Discussion, feedback, next meeting and closing



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1. Introduction

The Statistical Scraping Interest Group (SSIG) was established as a continuation of the network developed during the Web Intelligence Network (WIN) project. It connects national and international initiatives concerned with the use of web data and web scraping for statistical production.

The group and its meetings are co-organized by Statistics Netherlands and Statistics Austria. The third SSIG meeting took place on 16–17 September 2026 in Porto, Portugal. The SSIG continues to function as an informal community of practice in which participants share implementation experience, methodological questions and reusable tools, with the longer-term objective of strengthening the methodological foundations of statistical scraping.

The programme of SSIG3 combined updates from national statistical institutes and Eurostat with research contributions on web-derived economic classifications and automated scraper generation. Particular attention was given to reusable software, the integration of web data with business registers, the use of large language models, and the validation and maintainability of scraping pipelines.

2. Participants

FIRSTNAME	LASTNAME	Institution	Country
Alexander	Kowarik	Statistics Austria	AT
Christian	Salwiczek	Statistik Nord	DE
Olav	ten Bosch	Statistics Netherlands	NL
Dominik	Blatt	Statistics Netherlands	NL
Luuk	Haarman	Statistics Netherlands	NL
Luca	Gramaglia	European Commission - Eurostat	EC
José	Fragoso Santos	Instituto Superior Técnico Lisbon	PT
Dennis	Pipenbring	Statistics Denmark	DK
Bernhard	Meindl	Statistics Austria	AT
Emmanouil	Tranos	University of Bristol	UK
Jacek	Maślankowski	Statistics Poland	PL

3.



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3. Agenda

16 September at 9:30

9:00 - 9:30 Registration open

9:30 - 10:00 Start, welcome, SSIG in brief, who is here — Olav ten Bosch, Statistics Netherlands & Alexander Kowarik, Statistics Austria

10:00 - 10:45 Statistical scraping Use-cases in Austria: WEB-FOSS-AT / R-package taRantula — Alexander Kowarik, Bernhard Meindl, Statistics Austria, AT

10:45 - 11:15 Coffee Break

11:15 - 12:00 Statistical Scraping in Netherlands: WEB-FOSS-NL — Olav ten Bosch, Dominik Blatt, Luuk Haarman, Statistics Netherlands, NL

12:00 - 13:30 Lunch break

13:30 - 14:15 Integrating Domain Data into the Danish Statistical Business Register — Dennis Pipenbring, Statistics Denmark, DK

14:15 - 15:00 An Atlas of Economic Activities in the UK: A fine-grained typology of economic activities from open, large-scale Web Archive data — Emmanouil Tranos, University of Bristol, UK

15:00 - 15:30 Coffee Break

15:30 - 16:15 Job title cleaning pipeline — Luca Gramaglia, Eurostat

16:15 - 16:45 Discussion, closing first day (all)

17 September at 9:30

9:00 - 9:30 Registration open

9:30 - 10:15 Recent Achievements in Project Implementation at Statistics Poland — Jacek Maślankowski, Statistics Poland, PL

10:15 - 10:45 Coffee Break

10:45 - 11:30 WebCAP: Web Data Collection Via Automated Program Synthesis — José Fragoso Santos, Instituto Superior Técnico, PT

11:30 - 12:00 Discussion, feedback, next meeting, closing



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4. Meeting summary

1. Start, welcome, SSIG in brief, who is here (Olav ten Bosch, Alexander Kowarik)

The opening presentation set the context for the third SSIG meeting and revisited the motivation for using web data in official statistics. The main motivations highlighted were reducing survey burden by obtaining information from the web, increasing timeliness, developing indicators that cannot easily be measured using traditional sources, improving administrative and register data, and discovering missing entities.

The introduction also recalled work carried out under the Web Intelligence Network (WIN) and emphasized that major methodological challenges remain. In particular, web data need a sound framework for design, validation and estimation before they can support robust statistical conclusions. SSIG provides a forum for comparing approaches and developing this methodological perspective across institutions.

2. Statistical scraping Use-cases in Austria: WEB-FOSS-AT / R-package taRantula (Bernhard Meindl, Alexander Kowarik)

The Austrian contribution provided an update on WEB-FOSS-AT and on the development of tarantula (<https://github.com/statistikat/tarantula>), an R package for selective and reproducible web scraping. The work builds on use cases in official statistics where websites are linked to statistical units and only relevant pages or information are collected, rather than indiscriminately downloading complete websites.

The presentation revisited Statistics Austria use cases, including the collection of price information from enterprise websites and the use of web content to support economic-activity classification. These applications require a workflow that combines URL finding, focused crawling, content extraction and downstream classification, while retaining links to the underlying statistical units and metadata.

taRantula is designed to turn these steps into a reusable workflow. The presentation showed components for managing URLs and candidate pages, carrying out searches and validation, running parallel browser-based scraping jobs, storing intermediate results, and retaining logs and links for later inspection. A central design objective is to make the scraping process configurable and reproducible, while allowing project-specific extraction logic to be added where necessary.

The contribution also highlighted practical limitations that continue to shape statistical scraping: websites differ substantially in structure, search engines and site navigation are imperfect sources for URL discovery, browser automation is resource-intensive, and robust retry, validation and logging mechanisms are necessary for production use. The package therefore focuses not only on extraction itself but on orchestration and traceability of the complete scraping process.

3. Statistical Scraping in Netherlands: WEB-FOSS-NL (Olav ten Bosch, Dominik Blatt, Luuk Haarman)

Statistics Netherlands presented recent progress in WEB-FOSS-NL, an open-source and modular workflow for identifying Online Job Advertisements (OJAs) on company websites and comparing the results with the Dutch quarterly Job Vacancy Survey. The workflow combines URL finding, focused scraping and classification. Recent technical improvements include asynchronous and parallel scraping with Scrapy, Playwright support for dynamic content, improved sitemap handling, controlled domain jumps and the use of schema.org JobPosting markup as an additional indicator for vacancy pages.

(<https://github.com/SNStatComp/urlfinding>, <https://github.com/SNStatComp/webfocusedscrape>)



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A second test run covered 5,933 companies from the Job Vacancy Survey sample in health and social work and wholesale and retail trade. URL finding provided 4,039 unique base URLs and around 120,000 pages were fetched. The test highlighted practical issues such as low-quality base URLs, duplicate downloads and difficulties maintaining the link between vacancies and the originating company when following external domains. Schema.org markup proved useful when available, but was too incomplete to replace a classifier: only around 16% of scraped pages contained the indicator, although it was present for roughly half of the identified OJAs.

Future work will improve both scraping and classification, including additional labelled data, model fine-tuning and possible cascading approaches combining rules, embedding-based classifiers and LLM prompting. An Open Web Index experiment also showed promising potential for enriching the Statistical Business Register, although data quality and scalability still need further assessment.

4. Integrating Domain Data into the Danish Statistical Business Register (Dennis Pipenbring)

The Statistics Denmark presentation focused on how domain and webpage information can be represented and linked to units in the Statistical Business Register. A key distinction in the proposed terminology is that webpages are subsets of domains: webpages are the objects that are scraped, while domains provide the broader registration and ownership context.

The presentation illustrated why a simple one-to-one link between a business and a domain is often insufficient. A domain can be accessible or inaccessible, can serve an alternative purpose or act as an inactive placeholder, and can refer to business segments or market segments rather than directly to the legal unit. The legal owner of a domain may also differ from the beneficial owner or from the company to which the webpage content refers.

The proposed data model therefore separates received domain-register data from information extracted from the web and introduces explicit relations between register units, domains and webpages. It also allows additional properties of websites to be represented, such as the presence of webshop, booking, login or social-media functionality. The contribution stressed that a richer data model is required if domain data are to become a stable part of business-register infrastructure rather than a project-specific auxiliary variable.

5. An Atlas of Economic Activities in the UK: A fine-grained typology of economic activities from open, large-scale Web Archive data (Emmanouil Tranos et al.)

The University of Bristol contribution presented an open, web-derived and LLM-assisted typology of UK economic activity. The motivation is that standard industrial classifications are necessarily coarse, update slowly and may become stale when firms change activity, whereas business websites describe products, services and specialisations in firms' own words.

The Atlas processes large-scale Common Crawl data for 2021 and 2024. The pipeline filters and geocodes web content, uses an LLM to clean and summarise activity-related text, embeds and clusters those summaries, induces a taxonomy with TnT-LLM, and then semantically deduplicates and aligns labels across years. The final output contains 5,102 unified activity labels and covers more than 823,000 business websites across the two time points.



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Validation was carried out in several ways. Geographic checks found coverage across all 363 UK local authorities. Benchmarking against FAME/SIC records showed broad correspondence but also clear sectoral differences in web visibility, particularly for primary and secondary industries. Case studies demonstrated that the web-derived labels can identify more specific or emerging activities, for example in life sciences and clean energy, where conventional SIC codes may be too broad or provide no dedicated category.

The authors emphasized that the Atlas is not a census and should not be treated as ground truth. It covers web-visible businesses, has sector-specific coverage bias, and its labels do not form a strict hierarchy. Its intended role is therefore to complement existing classifications with a more granular, dynamic and geographically detailed view of economic activity.

6. Job title cleaning pipeline (Luca Gramaglia)

Eurostat presented a pipeline for cleaning job titles in Online Job Advertisements (OJA). Job titles are useful for identifying occupations and monitoring how occupational terminology evolves, but raw titles frequently contain location names, employer names, working-time information, recruitment wording and other context that is not part of the occupation itself.

The pipeline first extracts distinct raw titles and applies deterministic canonical normalization such as lowercasing and collapsing repeated punctuation and spaces. Distinct tokens are then extracted and classified with an LLM according to their occupational relevance: core occupational terms, occupational specialisations or domains, potentially useful modifiers, non-occupational context, or unclear/context-dependent tokens. Tokens classified as irrelevant are removed, after which cleaned titles are aggregated and ranked.

For each ISCO occupation at level 3, the 50 most frequent cleaned titles were compiled for every language covered by Eurostat's OJA collection. The lists are intended for expert review through the LAMAS working group, especially to identify occupations where misclassification remains prevalent. Planned improvements include context-sensitive handling of ambiguous tokens, multilingual titles and semantic merging or deduplication of very similar cleaned titles. The presentation also reported ongoing work on personally identifiable information (PII) redaction in OJA data using the PRESIDIO framework.

7. Recent Achievements in Project Implementation at Statistics Poland (Jacek Maślankowski)

Statistics Poland reported on progress in a project integrating web data into the statistical production system. The project is organized around four work packages: project governance and capacity building, integration of Online Job Advertisements (OJA), Online Business Enterprise Characteristics (OBEC), and a secure implementation architecture. A common requirement is that the substantive work packages use the Web Intelligence Hub (WIH) environment.

For OJA data, the objective is to identify skills associated with newly created jobs. Eurostat OJA data are linked to the Polish Business Register using company names and auxiliary variables and are then connected, inside the secured operational microdata environment, to social-insurance data identifying newly created jobs. A two-stage company-name matching procedure retrieves candidate companies and selects the best match using a hybrid of token-set ratio, Levenshtein distance and Jaro-Winkler similarity. The reported validation had produced 27,820 unambiguous matches, with roughly 8,000 ambiguous cases left for manual verification; the methodology had also been applied at scale to monthly 2024 data.



The presentation also covered implementation of OBEC data and the secure architecture required to work with linked register and web data. The overall approach separates exploratory web-data processing from protected microdata linkage and releases only appropriate aggregated outputs from the closed environment. This illustrates how methodological development, shared European web-data infrastructure and national security requirements can be combined in a production-oriented workflow.

8. WebCAP: Web Data Collection Via Automated Program Synthesis (José Fragoso Santos)

WebCAP aims to automate the construction and deployment of web scrapers for public-sector use cases, including retail prices, housing prices and employment offers. The original architecture separated navigation-code generation from extraction-query generation and provided a deployment layer for repeated execution, storage, APIs and dashboards.

Evaluation on a broad set of candidate websites exposed the limits of a rigid synthesis pipeline. Of 388 collected candidate websites, 198 housing and retail sites were actually tested after excluding sites blocked by captchas, Cloudflare, HTTP errors or similar restrictions. Only 21 working scrapers were obtained, corresponding to a success rate of about 10.6% with a human in the loop. Failures were highly site-specific: search fields acted as dropdowns, navigation landed on unexpected pages, pagination used carousels instead of links, or target values were split across multiple elements.

A second experiment showed that general-purpose coding agents could write site-specific Playwright scrapers substantially more successfully: 31 of 41 tested sites were solved in the reported experiment. This shifted the project focus. The central challenge is no longer generating scraper code, but establishing whether a generated scraper is correct and remains correct as websites change.

WebCAP 2.0 therefore proposes instrumenting scraper execution to produce a machine-readable trace, expressing the intended navigation and extraction behavior in a small human-reviewable specification language, and checking the execution trace against that specification. A failed conformance check produces a localized counterexample that can be fed back into repair. The specification can then be rerun during scheduled collection so that site redesigns become detectable violations rather than silent degradation. The presentation stressed, however, that technical conformance does not by itself establish statistical representativeness.

9. Discussion, feedback, next meeting and closing

The discussion covered several practical and methodological issues related to the future of statistical web scraping. One important topic was URL finding. The use of Google Custom Search Engine will no longer be possible after the end of 2026, making an alternative necessary. Brave Search appears to offer a close replacement and could potentially be integrated into existing workflows with limited changes. However, its retrieval quality still needs to be evaluated thoroughly. The Open Web Index was also discussed as a possible alternative. While its use appears technically feasible, there are open questions concerning both the long-term sustainability of the initiative and the quality of its results for identifying the correct websites of statistical units.

The use of agentic AI for developing and operating web scrapers was discussed extensively, but no clear preferred approach emerged. One option is to use AI agents to generate conventional code-based scrapers, which provides reproducible and inspectable programs that can subsequently be executed



independently of the AI system. At the same time, there are potential advantages in allowing an AI agent to interact with websites more dynamically, for example by using Playwright as a browsing tool rather than generating a scraper that depends on fixed XPath or other predefined selectors. Such an approach could offer greater flexibility and, in principle, allow the agent to adapt automatically when the structure of a website changes. Further experimentation is needed to better understand the trade-offs between robustness, flexibility, reproducibility, maintainability, and computational cost.

The next SSIG meeting will take place on 20–21 April 2027 in Copenhagen and will be hosted by Statistics Denmark.



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